

# Learning objectives

In this lesson you will learn to:

- apply logical left and right shifts to binary integers
- use logical binary shifts to multiply and divide unsigned binary integers by powers of 2
- explain why a number may be less precise after a binary shift right has been applied.

For more information about this topic and additional student activities see Topic 2 of the student book

# Binary arithmetic

You've learned how to add in binary, but what about other operations?

Multiplication and division can be achieved using **shifting**.

There are two forms of shifting:

**logical** and  
**arithmetic.**

# Logical left shifting

Shifts are known as 'bitwise operations' because they operate on each bit in a pattern, one at a time.

Left-shift      0 0 0 1 0 1 1 1      (decimal 23)



= 0 0 1 0 1 1 1 0 ← 0      (decimal 46)

The MSB is shifted left, out of the pattern.

Each remaining bit is shifted left to the next location.

A new 0 is placed into the vacated location on the right (LSB).

This has the effect of multiplying by 2.

NB **MSB – most significant bit**

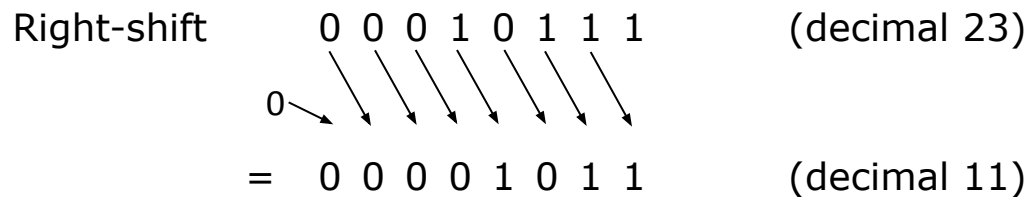
**LSB – least significant bit**

# Logical right shifting

The LSB is shifted right, out of the pattern.

A 0 is **always** inserted into the vacated MSB.

This has the effect of dividing by 2.



This causes a loss of precision.

23 divided by 2 should be 11.5, but we can only currently represent integers.

The 0.5 is lost.

This means that the result is less precise.

The more right shifts you perform, the more precision you may lose.

## Shifting more than once

What effect would shifting three times have?

How would you work out the effect?

You know that one shift divides or multiplies by two.

You should remember that binary is base two, which means that each column value is a multiple of two.

Shifting is the same, each shift is another divide or multiply by two.

For example, shifting three times would be the same as dividing or multiplying by 8.

# Wrap up: so far you have learned how to...

- ✓ Apply logical left and right shifts to binary integers.
  - Shifting left one position – a new 0 is shifted in on the right (LSB).
  - Shifting right one position – a new 0 is shifted in on the left (MSB).
- ✓ Use logical binary shifts to multiply and divide unsigned binary integers by powers of 2.
  - Each shift is another power of two.
- ✓ Explain why a number may be less precise after a binary shift right has been applied
  - Decimal places are lost when you divide by shifting.